ASHREI TECH | NASTP

Python for AI

Course Overview

- **Duration:** 3 Months (13 Weeks)
- **Classes per Week:** 5

Week-by-Week Curriculum

Week	Module Title	Day	Learning Units
Week 1	**Core Language Foundations**	Day 1	Orientation & Python Basics
		Day 2	Conditionals & Reading Errors
		Day 3	Loops
		Day 4	Lists & Tuples, In Depth
		Day 5	Dictionaries & Sets + Weekly Skill Checkpoint
Week 2	**Functions, OOP & Robust Code**	Day 1	Functions I & II + Comprehensions
		Day 2	Lambda, Sorting & Higher-Order Functions
		Day 3	OOP I, Classes & Objects
		Day 4	OOP II, Inheritance & Polymorphism
		Day 5	Exception & File Handling + Weekly Skill Checkpoint
Week 3	**Python Data Libraries & Exploratory Data Analysis**	Day 1	NumPy
		Day 2	Pandas I
		Day 3	Pandas II
		Day 4	Matplotlib & Seaborn
		Day 5	Weekly Skill Checkpoint
Week 4	**Classical Machine Learning & Neural Networks**	Day 1	The ML Workflow & Linear Regression
		Day 2	Classification Models
		Day 3	Tree-Based Models & Evaluation
		Day 4	Neural Networks, Conceptually
		Day 5	Neural Networks in PyTorch + Weekly Skill Checkpoint
Week 5	**Deep Learning, the Bridge to LLMs & Practical NLP**	Day 1	CNNs, Lightly
		Day 2	Deep Learning Mini-Project & Checkpoint
		Day 3	Bridge to LLMs
		Day 4	LLM APIs & Efficient Prompting
		Day 5	NLP Fundamentals & a Classical Chatbot
Week 6	**LLM Applications, Transformers & the Start of RAG**	Day 1	Month 2 Project Kickoff
		Day 2	Building a Simple Chatbot (LLM-Powered)
		Day 3	How LLMs Actually Work: RNNs & Transformers + Checkpoint
		Day 4	Embeddings & Vector Databases
		Day 5	Document Loading & Chunking
Week 7	**RAG & the Start of Agents**	Day 1	Building the Retrieval Pipeline
		Day 2	Improving RAG Quality

Week	Module Title	Day	Learning Units
		Day 3	Weekly Skill Checkpoint, RAG Pipelines
		Day 4	What Is an Agent?
		Day 5	Tool Calling & MCP Fundamentals + Weekly Skill Checkpoint
Week 8	**Agents & MCP**	Day 1	Build Your Own MCP Server
		Day 2	LangGraph: Stateful Agents
		Day 3	Weekly Skill Checkpoint, Agents & MCP
		Day 4	Feature Completion
		Day 5	Testing & Edge Cases
Week 9	**Demo Day & Git/API Foundations**	Day 1	Interface Polish
		Day 2	Documentation & Polish
		Day 3	Demo Day
		Day 4	Environments, Project Structure & Git
		Day 5	APIs, Client-Server & FastAPI
Week 10	**Servers, APIs & Deployment**	Day 1	Request Bodies, Validation & CORS
		Day 2	Wrap Your Agent in an API
		Day 3	Containerization
		Day 4	Cloud Deployment
		Day 5	Weekly Skill Checkpoint + Deployment
Week 11	**Advanced Agents: Evaluation, Orchestration & Voice**	Day 1	Agent Evaluations
		Day 2	Guardrails, Safety & Observability
		Day 3	Multi-Agent Orchestration
		Day 4	Voice & External Integrations
		Day 5	Weekly Skill Checkpoint, Advanced Agents
Week 12	**Portfolio & Freelance**	Day 1	GitHub Portfolio & LinkedIn
		Day 2	Freelancing on Upwork & Fiverr
		Day 3	Pakistan's AI Job Market & Resume
		Day 4	Technical Interview Preparation
		Day 5	Weekly Skill Checkpoint
Week 13	**Assessment and Final Project**	Day 1	Buffer & Instructor Catch-Up
		Day 2	Buffer & Instructor Catch-Up
		Day 3	Presentation Rehearsal
		Day 4	Final Presentations, Day 1
		Day 5	Final Presentations, Day 2